Shubham Farera

Senior Software Engineer (SDE-3) / Distributed Systems & Backend Architect / Java, Spring Boot & Kafka Specialist

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Senior Software Engineer with 8.5+ years of experience architecting, designing, and scaling distributed backend systems and high-throughput microservices. Deep expertise in High-Level Design (HLD) and Low-Level Design (LLD), applying SOLID principles, GoF design patterns, Data Structures & Algorithms, and clean architecture to build extensible, fault-tolerant enterprise applications. Proven track record in distributed systems engineering: designing event-driven architectures using Apache Kafka, implementing distributed transaction patterns (Saga pattern, Transactional Outbox), low-latency search with Elasticsearch, and optimizing database tiers through sharding and master-slave replication. Skilled in navigating CAP and PACELC trade-offs to build highly available, low-latency platforms across Healthcare, FinTech, and Logistics.

Executive Highlights

Scope & Impact

- 8.5+ years of experience architecting, designing, and delivering high-throughput, fault-tolerant distributed backend systems.
- Proven SDE-2/SDE-3 level expertise in High-Level Design (HLD), Low-Level Design (LLD), Data Structures & Algorithms, and clean architecture.

Architecture & Design

- Architected distributed event-driven microservices utilizing Apache Kafka for asynchronous communication, pub-sub messaging, and high throughput.
- Implemented the Saga Pattern (Orchestration & Choreography) and Outbox Pattern to maintain data consistency in distributed transactions.

- Designed and scaled event-driven architectures with Apache Kafka, processing mission-critical transactions across enterprise domains.
- Spearheaded distributed transaction management with the Saga pattern and Transactional Outbox pattern across microservices.
- Enforced rigorous code craftsmanship: SOLID principles, GoF design patterns, CodeQL static analysis, and SonarQube quality gates.
- Captain of state-level CBSE Football team and represented institute at National Level Table Tennis tournament.
- Engineered robust Low-Level Designs (LLD) strictly adhering to SOLID principles, Design Patterns (Factory, Strategy, Observer, Decorator, State), and clean code.
- Designed resilient data tiers incorporating Master-Slave read replication, database sharding/partitioning, and Consistent Hashing.
- Integrated Elasticsearch clusters for high-performance distributed search, custom indexing, and typeahead lookups.
- Applied distributed systems fundamentals including CAP and PACELC theorems, ZooKeeper coordination, and multi-tier Redis caching.

Key Platforms

- Distributed Event-Driven Insurance Platform: High-throughput Spring Boot microservices backed by Apache Kafka and the Saga pattern.
- Elasticsearch Search & Indexing Engine: Low-latency search and typeahead retrieval reducing query latencies by over 60%.
- Asset Rebalancing & Investment Platform: Secure backend services automating portfolio rebalancing and financial calculations.
- Logistics Order Management Engine: Multi-stage order lifecycle processing utilizing design patterns and optimized SQL data layers.
- Cloud & Security Infrastructure: Microsoft Azure (AZ-900 certified), automated Jenkins CI/CD, CodeQL, and SonarQube quality gates.

Skills

Java (8/11/17) | Data Structures & Algorithms (DSA) | High-Level Design (HLD) | Low-Level Design (LLD) | SOLID Principles | Design Patterns (GoF) | Spring Boot & Spring Cloud | Microservices Architecture | Apache Kafka | Distributed Transactions (Saga Pattern) | Elasticsearch | Redis & Distributed Caching | Database Sharding & Replication | Consistent Hashing | ZooKeeper | SQL & NoSQL Databases | MySQL & Oracle | RESTful APIs | JPA & Hibernate | Azure (AZ-900 Certified) | Jenkins (CI/CD) | CodeQL & SonarQube | Git & GitHub | Agile / Scrum

Experience

IT Analyst (Senior Backend Engineer)

Tata Consultancy Services · Oct 2023 – Present

- Architected and developed scalable, high-throughput microservices using Java, Spring Boot, and Apache Kafka for event-driven asynchronous processing of healthcare claims and policy workflows.
- Implemented distributed transactions using the Saga Pattern (Orchestration and Choreography), maintaining cross-service data consistency without two-phase commit overhead.
- Engineered low-latency search and policy retrieval services using Elasticsearch, designing index mappings and typeahead search capabilities that reduced search latency by over 60%.
- Applied Low-Level Design (LLD) and SOLID principles with GoF design patterns (Strategy, Factory, Observer, Decorator) to build modular, clean, and extensible service components with 85%+ test coverage.
- Optimized database performance by designing master-slave read replication architectures, database sharding/partitioning strategies, and multi-level Redis caching to eliminate bottlenecks.
- Designed fault-tolerant architectures utilizing circuit breakers (Resilience4j), rate limiting, and graceful degradation, applying CAP and PACELC trade-offs for high availability and partition tolerance.
- Managed CI/CD deployment automation using Jenkins, ensuring zero critical vulnerabilities via CodeQL static analysis and SonarQube quality standards.

Application Development Senior Analyst

Accenture · Sept 2021 – Aug 2023

- Singapore-based Asset Rebalancing and Investment System: Developed backend microservices for automated portfolio rebalancing workflows and asset allocation calculations.
- Implemented secure, idempotent RESTful APIs integrating with third-party financial institutions and payment networks.
- Optimized complex SQL queries, transaction management, and Hibernate ORM mappings for high-volume financial data processing.
- Maintained zero-defect code quality standards and led vulnerability remediations using SonarQube.

Application Development Analyst

Accenture · May 2020 – Aug 2021

- Australia-based Logistics Order Management System: Designed and developed core order processing and tracking REST APIs using Spring Boot and JPA.
- Applied State and Strategy design patterns to manage complex, multi-stage order lifecycles and tracking state transitions.
- Enhanced database indexing and query execution plans to improve throughput and support growing shipping volumes.

Associate Software Engineer

Accenture · Feb 2018 – Apr 2020

- US-based Healthcare and Child Support Project: Built strong foundational engineering expertise in Core Java, Data Structures, OOP, and clean code principles.
- Developed internal service modules, data transformation utilities, and batch jobs for enterprise client workflows.
- Executed unit and integration tests using JUnit and Mockito within Agile development sprints.

Education

Bachelor of Engineering – Information Technology, SGSITS Indore (Indore, MP) (2013 – 2017) - CGPA 7.1

12th Standard – PCM, Vivekanand Vidya Vihar, Maral Sarovar (CBSE) (2011 – 2012) - 87%

10th Standard, Vivekanand Vidya Vihar, Maral Sarovar (CBSE) (2009 – 2010) - CGPA 9.0